

Part A: Linux Installation & Verification

Screenshots are attached with this file in the screenshot section.

Part B: Basic Navigation Commands

```
sneha@kali: ~
$ pwd
/root/.sneha
$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
$ ls -la
total 180
drwxr-xr-x 17 sneha sneha 4096 Jun 12 08:23 .
drwxr-xr-x 3 root root 4096 Apr 3 05:38 ..
-rw-r--r-- 1 sneha sneha 228 Apr 3 05:38 .bash_logout
-rw-r--r-- 1 sneha sneha 5578 Apr 3 05:38 .bashrc
-rw-r--r-- 1 sneha sneha 3528 Apr 3 05:38 .bashrc.original
drwxr-xr-x 12 sneha sneha 4096 Apr 11 11:05 .cache
drwxr-xr-x 16 sneha sneha 4096 Apr 3 05:38 .config
drwxr-xr-x 2 sneha sneha 4096 Apr 3 05:44 Desktop
-rw-r--r-- 1 sneha sneha 35 Apr 3 05:44 .dirc
drwxr-xr-x 2 sneha sneha 4096 Apr 3 05:44 Documents
drwxr-xr-x 2 sneha sneha 4096 Apr 3 05:53 Downloads
-rw-r--r-- 1 sneha sneha 11779 Apr 3 05:38 .face
-rwxrwxrwx 1 sneha sneha 5 Apr 3 05:38 .face.icon -> .face
drwxr-xr-x 1 sneha sneha 4096 Apr 3 05:44 .gpgg
-rw-r--r-- 1 sneha sneha 8 Apr 3 05:44 .ICAuthority
drwxr-xr-x 3 sneha sneha 4096 Apr 3 05:38 .java
drwxr-xr-x 5 sneha sneha 4096 Apr 3 05:44 .local
drwxr-xr-x 5 sneha sneha 4096 Apr 3 05:48 .mozilla
drwxr-xr-x 2 sneha sneha 4096 Apr 3 05:44 Music
drwxr-xr-x 2 sneha sneha 4096 Jun 12 08:17 Pictures
-rw-r--r-- 1 sneha sneha 887 Apr 3 05:38 .gprofile
drwxr-xr-x 2 sneha sneha 4096 Apr 3 05:44 Public
drwxr-xr-x 1 sneha sneha 4096 Apr 3 05:44 .ssh
-rw-r--r-- 1 sneha sneha 8 Apr 3 05:46 .sudo_as_admin_successful
drwxr-xr-x 2 sneha sneha 4096 Apr 3 05:44 Templates
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-clipboard-tyt7-control.pid
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-clipboard-tyt7-service.pid
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-draganddrop-tyt7-control.pid
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-draganddrop-tyt7-service.pid
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-hostversion-tyt7-control.pid
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-seamless-tyt7-control.pid
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-seamless-tyt7-service.pid
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-vmtoolsd-session-tyt7-control.pid
-rw-r--r-- 1 sneha sneha 5 Jun 12 08:02 .vboxclient-vmtoolsd-session-tyt7-service.pid
drwxr-xr-x 1 sneha sneha 4096 Apr 12 05:14 Videos
-rw-r--r-- 1 sneha sneha 49 Jun 12 08:02 .Xauthority
-rw-r--r-- 1 sneha sneha 7013 Jun 12 08:18 .xsession-errors
-rw-r--r-- 1 sneha sneha 7258 Apr 3 05:38 .xsession-errors.pid
-rw-r--r-- 1 sneha sneha 336 Apr 3 05:38 .zprofile
```

```
sneha@kali: ~/Documents
$ cd Documents
$ history
1 hostnmecl
2 hostnmecl
3 pwd
4 ls
5 ls -la
6 pwd
7 ls
8 ls -la
9 cd
10 cd Documents
11 clear
12 history
13 whoami
14 hostnmecl
15 hostnmecl
16 ls
17 ls -la
18 clear
19 cd Documents
$ whoami
sneha
$ hostnmecl
kali
$
```

pwd – Displays the full path of the current working directory, showing your current location in the Linux file system.

ls – Lists the files and directories present in the current directory.

ls -la – Displays a detailed list of all files and directories, including hidden files, along with permissions, owner, size, and modification date.

cd – Changes the current working directory to another directory specified by the user.

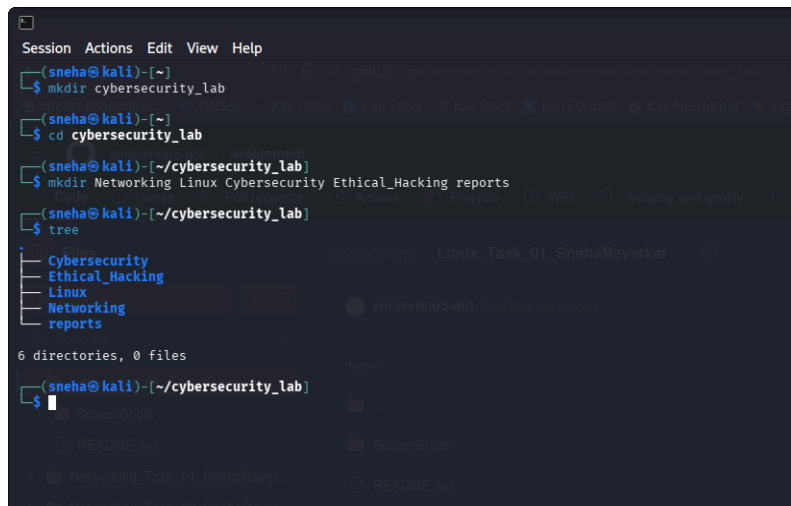
clear – Clears all previous text from the terminal screen, providing a clean workspace.

history – Shows a list of commands that have been previously executed in the terminal.

whoami – Displays the username of the currently logged-in user.

hostname – Displays the name of the computer or system on the network.

Part C: Directory Management



```
Session Actions Edit View Help
(sneha@kali)-[~]
$ mkdir cybersecurity_lab
(sneha@kali)-[~]
$ cd cybersecurity_lab
(sneha@kali)-[~/cybersecurity_lab]
$ mkdir Networking Linux Cybersecurity Ethical_Hacking reports
(sneha@kali)-[~/cybersecurity_lab]
$ tree
.
├── cybersecurity
├── Ethical_Hacking
├── Linux
├── Networking
└── reports
6 directories, 0 files
(sneha@kali)-[~/cybersecurity_lab]
$
```

Part D: File Management

1. Created the main folder **CyberSecurity_Lab** and the subfolders **Networking**, **Linux**, **CyberSecurity**, **EthicalHacking**, and **Reports** using the **mkdir** command.
2. Navigated to the **CyberSecurity** folder inside **CyberSecurity_Lab** using the **cd** command.
3. Created three files named **notes.txt**, **commands.txt**, and **report.txt** using the **touch** command.
4. Verified that the files were created successfully using the **ls** command.
5. Copied **notes.txt** to a new file named **notes1.txt** using the **cp** command.
6. Renamed **report.txt** to **renamed.txt** using the **mv** command.
7. Moved **commands.txt** from the **CyberSecurity** folder to the **Linux** folder using the **mv** command.
8. Deleted the copied file **notes1.txt** using the **rm** command.
9. Checked the final contents of the folders using the **ls** command to confirm that the operations were completed successfully.

```
Session Actions Edit View Help
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ touch notes.txt commands.txt report.txt
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ ls
commands.txt  notes.txt  report.txt
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ cp notes.txt notel.txt
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ ls
commands.txt  notel.txt  notes.txt  report.txt
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ mv report.txt renamed.txt
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ ls
commands.txt  notel.txt  notes.txt  renamed.txt
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ mv notel.txt
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ ls
commands.txt  notes.txt  renamed.txt
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ mv commands.txt ../Linux
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ cd ..
(sneha@kali) ~/cybersecurity_lab
$ cd Linux
(sneha@kali) ~/cybersecurity_lab/Linux
$ ls
commands.txt
(sneha@kali) ~/cybersecurity_lab/Linux
$ cd ..
(sneha@kali) ~/cybersecurity_lab
$ cd Cybersecurity
(sneha@kali) ~/cybersecurity_lab/Cybersecurity
$ ls
```

1. **touch** – Creates a new empty file. If the file already exists, it updates the file's timestamp.
2. **cp** – Copies a file or directory from one location to another.
3. **mv** – Moves a file or directory to a different location. It can also be used to rename files and directories.
4. **rm** – Deletes a file or directory from the system.

Part E: System Information Collection

```
Session Actions Edit View Help
(sneha@kali) [~]
$ uname -a
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64 GNU/Linux
(sneha@kali) [~]
$ hostname
kali
(sneha@kali) [~]
$ whoami
sneha
(sneha@kali) [~]
$ date
Fri Jun 12 09:13:01 AM EDT 2026
(sneha@kali) [~]
$ uptime
09:13:06 up 1:10, 1 user, load average: 0.95, 0.99, 0.80
(sneha@kali) [~]
$ pwd
/home/sneha
(sneha@kali) [~]
```

- Kernel Version:-6.18.12+kali-amd64
- Username:-Kali
- Current Directory:-/home/sneha
- Current Date and Time :-Fri Jun 12 09:13:01 AM EDT 2026

Part F: Linux Research Activity

1. What is Linux?

Linux is an operating system that helps a computer work properly. It manages hardware, software, files, and programs. Linux is free to use and is popular because it is secure and reliable.

2. Why is Linux important in Cyber Security?

Linux is important in Cyber Security because many security tools run on Linux. It allows users to perform security testing, network analysis, and system monitoring more effectively. Many servers also use Linux, so learning Linux is important for cybersecurity professionals.

3. Difference between Linux and Windows

Linux is a free and open-source operating system, while Windows is developed by Microsoft and requires a license. Linux provides more customization and is widely used by programmers and cybersecurity experts. Windows is generally easier for beginners and is commonly used on personal computers.

4. What is a Linux Distribution?

A Linux Distribution is a version of Linux that includes the operating system along with useful software and tools. Different distributions are designed for different purposes. Examples include Kali Linux, Ubuntu, and Parrot OS.

5. Why do Ethical Hackers prefer Linux-based operating systems?

Ethical hackers prefer Linux because it provides powerful security tools, better control over the system, and strong command-line support. Many penetration-testing and cybersecurity tools are available on Linux, making it a preferred choice for ethical hacking and security research.